

Final Project Guidelines: Introduction to Data Science

Objective

The objective of this final project is to apply the complete Data Science life cycle — from data collection to model development, evaluation, and interpretation — using R on real-world datasets.

This project emphasizes automation, reproducibility, and breadth of understanding:

- You must use real, standard datasets from reliable sources or collected from the web.
- Your code must be dataset-independent as much as possible and runnable on any computer without modification.
- Each modeling task must be implemented in separate, well-structured R scripts.

Project Options

You must choose one of the following options:

Option 1 — Structured Datasets (Classification, Regression, and Clustering)

Option 2 — Text-Based Datasets (NLP Classification, Topic Modeling, and Clustering)

Option 1 — Structured Data Projects

Overview

In this option, you will implement three independent R scripts, each focusing on a different modeling task:

1. Classification
2. Regression
3. Clustering

Each project must use a separate R script. You may continue your midterm dataset for one of them if it fits the requirement.

Dataset Requirements

- Each dataset must be real-world and standard (e.g., Kaggle, UCI, data.gov).
- Minimum 500 rows per dataset.
- Each must include both categorical and numerical features.

- Datasets should represent different domains (e.g., health, marketing, education, finance, sports, etc.).

A. Data Collection (for each task)

- Provide dataset link, source, and short description in your code comments.
- Import data automatically (no hard-coded local paths).
- Ensure data loading code works without modification.

B. Data Understanding & Exploration

Perform standard EDA for each dataset:

- Display shape, data types, and summary statistics.
- Identify categorical and numerical features.
- Visualize data distributions and correlations (histograms, boxplots, scatterplots, heatmaps).
- Identify missing values, outliers, and relationships.

C. Data Preprocessing

Automate preprocessing for each dataset:

- Handle missing values and outliers.
- Encode categorical variables automatically.
- Normalize or standardize numerical variables.
- Apply transformations to reduce skewness where needed.
- Apply feature engineering

D. Modeling

Implement three separate models across three different datasets:

Task	Example Models	Output Type
Classification	Decision Tree	Categorical label
Regression	Linear Regression	Continuous value
Clustering	K-Means	Cluster assignments

E. Model Evaluation & Interpretation

- Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix
- Regression: RMSE, MAE, R^2
- Clustering: Silhouette Score, Cluster Visualization

Provide:

- Clear output print statements and plots in your R code.
- Brief in-code comments explaining model interpretation.

F. Output and Code Files

You must submit three separate scripts:

1. C-G01-final-classification.R
2. C-G01-final-regression.R
3. C-G01-final-clustering.R

Each script must:

- Run independently and automatically
- Contain dataset loading, preprocessing, modeling, and evaluation steps
- Be fully commented and reproducible

Option 2 — Text-Based (NLP) Projects

Overview

In this option, you will work with unstructured text data and complete three distinct NLP projects:

1. Text Classification (e.g., Sentiment Analysis)
2. Topic Modeling
3. Text Clustering

Each must use a separate dataset and a separate R script.

Dataset Requirements

- Collect or use real-world text datasets (reviews, articles, blogs, tweets, etc.)
- Sources: Kaggle, Hugging Face, data.world, or **web scraping** via `rvest`, `httr`, or APIs
- Each dataset must have at least 500 documents

A. Data Collection

- Clearly mention data source, collection method, and domain in code comments.
- Provide the link or script used for data scraping or download.
- Use automated data import (Google Drive, GitHub, or web link).

B. Text Understanding & Exploration

- Show total documents, average text length, and vocabulary size.
- Display most frequent words and n-grams.
- Visualize word clouds, bar plots, and sentiment distribution.

C. Text Preprocessing

Apply a full NLP preprocessing pipeline:

- Convert text to lowercase.
- Remove punctuation, numbers, URLs, and stopwords.
- Perform tokenization, stemming, or lemmatization.
- Create Document-Term Matrix (DTM) or TF-IDF representation.

Ensure the code automatically detects text columns and applies preprocessing.

D. Modeling

Implement three NLP projects using different datasets:

Task	Example Models/Techniques	Output Type
Text Classification	Naïve Bayes or Logistic Regression	Sentiment/Category label
Topic Modeling	LDA (Latent Dirichlet Allocation)	Topics
Clustering	K-means	Grouped documents

E. Evaluation & Interpretation

- Classification: Accuracy, F1-score, Confusion Matrix
- Topic Modeling: Topic Coherence, Top Words Visualization
- Clustering: Silhouette Score, Cluster Visualization

F. Output and Code Files

You must submit three separate scripts:

1. C-G01-final-nlp-classification.R
2. C-G01-final-nlp-topicmodel.R
3. C-G01-final-nlp-clustering.R

Each script must:

- Run independently and automatically
- Contain full workflow (data import → preprocessing → modeling → evaluation)
- Be fully commented and reproducible

Submission Requirements

- No report is required.
- Submit all R scripts via MS Teams.
- Ensure each file runs successfully and produces the expected outputs and visualizations.

Evaluation Criteria (Total 50 Marks)

Component	Marks
Data Collection & Description	10
Data Exploration & Preprocessing	10
Modeling & Evaluation (All Three Tasks)	20
Insights & Interpretation (via comments & visuals)	5
Code Quality (Automation & Readability)	5
Total	50 marks

Submission & Viva

- Deadline: 3rd January 2026, 11:59 PM sharp (**no late submissions accepted**)
- Submission: Upload all R scripts to MS Teams
- Viva: Individual viva will be held in the last week of the final term, before the exam week.

Important Notes

- You may continue or change your midterm dataset.
- Use real, standard datasets (no toy or synthetic data).
- Each R script must be as dataset-independent as possible and runnable on any computer without modification.
- **Late submission: 10% deduction per day.**
- Creativity, clarity, and visualization quality will earn bonus credit.